Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

For a complex number Z, if Z - $1 + i + Z + i = 1,$ then the range of the principle argument of Z is (where principle arg$Z \in$ -π, π)

(A) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$

(B) $\left[\frac{\pi}{4}, \frac{\pi}{2}\right]$

(C) $\left[-\frac{\pi}{2}, -\frac{\pi}{4}\right]$

(D) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

Q.2

If $\left| \mathrm{z} - 4 + 3\mathrm{i} \leqslant 2\right.$, then the least and the greatest values of $|\mathrm{z}|$ are

(A) 3,7

(B) 3,9

(C) 4,7

(D) None of these

Q.3

For an arbitrary complex $\mathrm{z}, \mathrm{z} - 3i| \geq 2, \mathrm{Re}(\mathrm{z})| \leq 2, \mathrm{I} \leq |\mathrm{Im}(\mathrm{z})| \leq 5$, then area of region representing all possible positions of ' z' is equal to:

(A) $16 - 4\pi$

(B) $32 - 2\pi$

(C) $32 - 4\pi$

(D) $16 - 2\pi$

Q.4

The roots of the equation $1 + z + z^3 + z^4 = 0$ are represented by the vertices of

(A) a square

(B) an equilateral triangle

(C) a rhombus

@IITJEE_Advanced

(D) none of these

Q.5

Squre ABCD has its centre at the orgin. If A is the point z_1, then the centroid of the triangle ABC is

(A) $\pm i\frac{z_1}{3}$

(B) $\pm\frac{z_1}{3}$

(C) $\frac{z_1}{3}e^{\pm\pi/3}$

(D) $\frac{z_1}{3}e^{\pm x/6}$

Q.6

The locus of a point in the Argand plane that moves satisfying the equation $|z-1+i|-|z-2-i|=3$:

(A) is a circle with radius 3 and centre at $z=\frac{3}{2}$

(B) is an ellipse with its foci at $1-\mathrm{i}$ and $2+\mathrm{i}$ and major axis $=3$

(C) is a hyperbola with its foci at $1-\mathrm{i}$ and $2+\mathrm{i}$ and its transverse axis $=3$

(D) None of the above

Q.7

$\mathrm{z}, \omega\mathrm{z}$ ane $\bar{\omega}\mathrm{z}$ are the vertices of a triangle, then If the area of the triangle will be (where ω is cube root of unity) :

(A) $\frac{3|z|^2}{2}$

(B) $\frac{3\sqrt{3}|z|^2}{2}$

(C) $\frac{\sqrt{3}|z|^2}{2}$

(D) None of these

Q.8

Let $\lambda\in\mathbf{R}$. If the origin and the non real roots of $2\mathrm{z}^2+2\mathrm{z}+\lambda=0$ form the three vertices of an equilateral triangle in the argand plane. Then λ is

(A) 1

(B) $\frac{2}{3}$

(C) 2

(D) -1

@IITJEE_Advanced

Q.9

Let z lies on the circle centred at the origin. If area of the triangle whose vertices are $\mathrm{z}, \omega \mathrm{z}$ and $\mathrm{z}+\omega \mathrm{z}$, where ω is the cube root of unity is $4 \sqrt{3}$ sq. unit. Then radius of the circle is:

(A) 1 unit

(B) 2 units

(C) 4 units

(D) None of these

Q.10

The greatest and the least absolute value of $z+1$, where $|z+4| \leq 3$ are respectively

(A) 6 and 0

(B) 10 and 6

(C) 4 and 3

(D) None of these

Q.11

z_1 and z_2 are the roots of $3 z^2+3 z+b=0$. If $\mathrm{O}(0), A\left(z_1\right), B\left(z_2\right)$ form an equilateral triangle, then the value of b is

(A) -1

(B) 1

(C) 0

(D) does not exist

Q.12

Let $\mathrm{A}_0 \mathrm{~A}_1 \mathrm{~A}_2 \mathrm{~A}_3 \mathrm{~A}_4 \mathrm{~A}_5$ be a regular hexagon
inscribed in a circle of unit radius. Then the product of the lengths of the line segments $\mathrm{A}_0 \mathrm{~A}_1, \mathrm{~A}_0 \mathrm{~A}_2$ and $\mathrm{A}_0 \mathrm{~A}_4$ is

(A) $\frac{3}{4}$

(B) $3 \sqrt{3}$

(C) 3

(D) $\frac{3 \sqrt{3}}{2}$

Q.13

@IITJEE_Advanced

If centre of a regular hexagon is at origin and one of the vertex on argand diagram is $1 + 2i$, then its perimeter is

(A) $2\sqrt{5}$

(B) $6\sqrt{2}$

(C) $4\sqrt{5}$

(D) $6\sqrt{5}$

Q.14

If the point $z_1 = 1 + i$ where $i = \sqrt{-1}$ is the reflection of a point $z_2 = x+$ iy in the line $i\bar{z} - iz = 5$, then the point z_2 is

(A) $1 + 4\mathrm{i}$

(B) $4 + \mathrm{i}$

(C) $1 - \mathrm{i}$

(D) $-1 - \mathrm{i}$

Q.15

If $|z - 2| = \min\{|z - 1|, |z - 5|\}$, where z is a complex number, then

(A) $\mathrm{Re}(z) = \frac{3}{2}$

(B) $\mathrm{Re}(z) = \frac{7}{2}$

(C) $\mathrm{Re}(z) \in \left\{\frac{3}{2}, \frac{7}{2}\right\}$

(D) None of these

Answers & Solutions

Q.1 Answer:

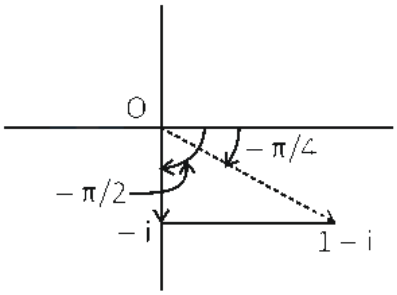
$-\frac{\pi}{2}, -\frac{\pi}{4}$

Solution:

$Z - 1 + i + Z + i = 1 - i - -i$

⇒ **locus of** Z **is the line segment joining** $-i$ **with** $1 - i$

@IITJEE_Advanced



⇒ **minimum arg** $Z = -\frac{\pi}{2}$ **& maximum arg** $Z = -\frac{\pi}{4}$

⇒ **principal arg** $Z \in -\frac{\pi}{2}, -\frac{\pi}{4}$

Q.2 Answer:

3,7

Solution:

Let $z = x+$ **iy**

Given, $|z-4+3i| \leq 2$

$\therefore |x+iy-4+3i| \leq 2$

or, $\sqrt{(x-4)^2+(y+3)^2} \leq 2$

or, $(x-4)^2+(y+3)^2 \leq 2^2$

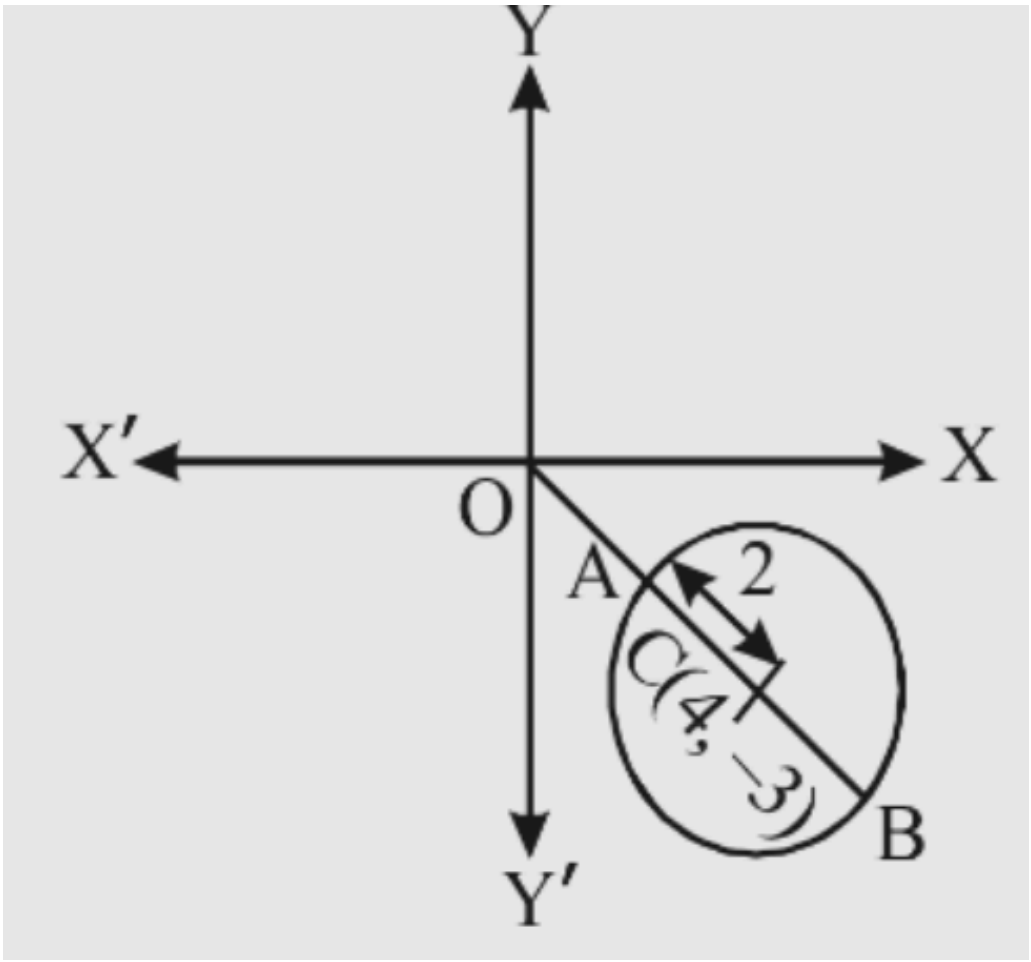
Thus, z lies in the interior or on the boundary of the circle whose centre is $(4,-3)$ **and radius is** 2.

Least value of $|z| = OA = OC - AC = 5-2 = 3$

Greatest value of $|$ **z.** $| = \mathrm{OB} = \mathrm{OC} + \mathrm{CB} = 5+2 = 7.$

@IITJEE_Advanced

Thus, $3 \quad 3 \leq z \left| \leq 7 \right.$



Q.3 Answer:

$32 - 4\pi$

Solution:

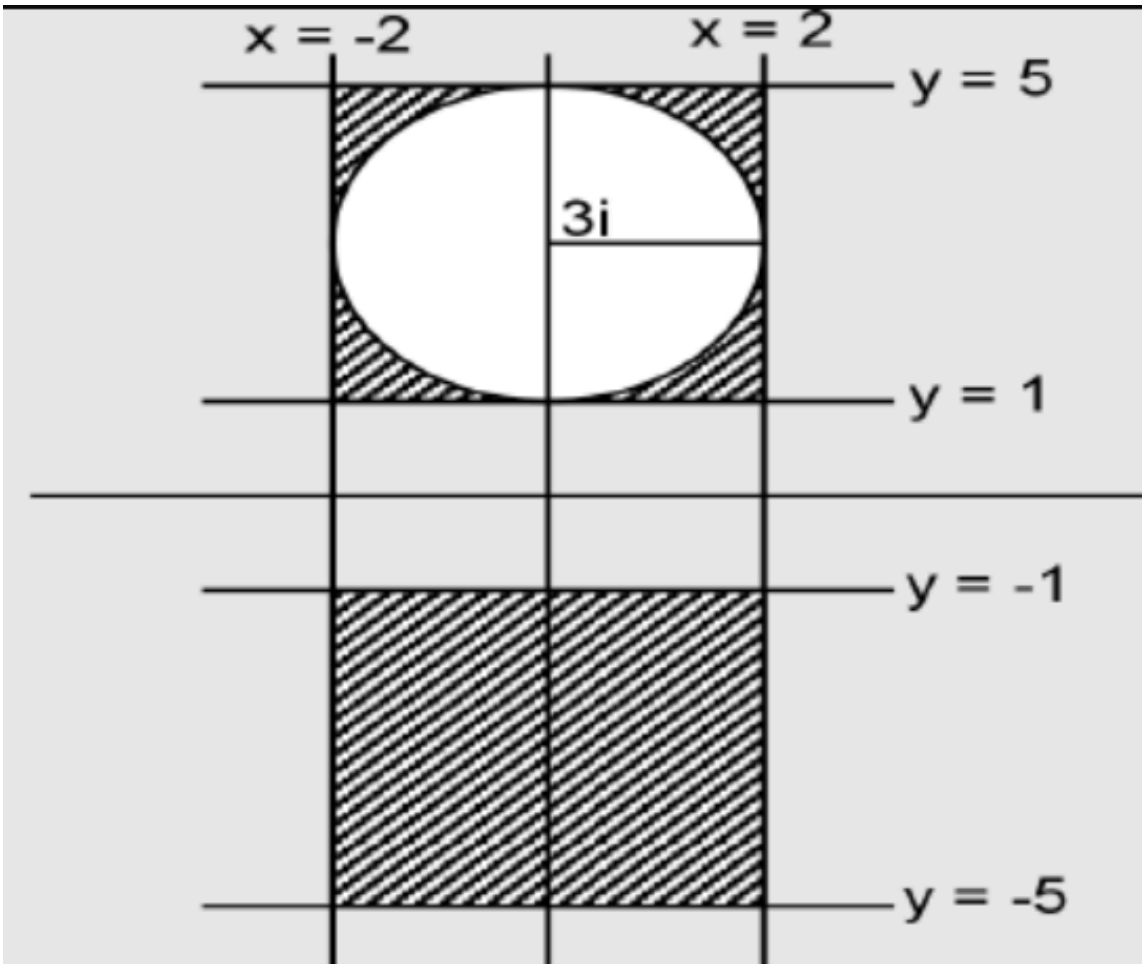
Required Area

$= 2$ **(Area of square)** $-($ **Area of circle** $)$

$= 2(4.4) - \pi \cdot 2^2$

@IITJEE_Advanced

$= 32 - 4\pi$



Q.4 Answer:

an equilateral triangle

Solution:

$1 + z + z^3 + z^4 = 0$

$\Rightarrow (1 + z) + z^3(1 + z) = 0$

$\Rightarrow \left(1 + z^3\right)(1 + z) = 0$

$z = -1$

$z^3 = -1$

$z = -1, -\omega, -\omega^2$ **are cube roots of unity**

So these are the vertices of an equilateral triangle.

Q.5 Answer:

$\pm i\frac{z_1}{3}$

Solution:

$\mathrm{OA} = \mathrm{OB} = \mathrm{OC}$. **Let** $\mathrm{B}\left(z_2\right)$ **and** $\mathrm{C}\left(z_3\right)$

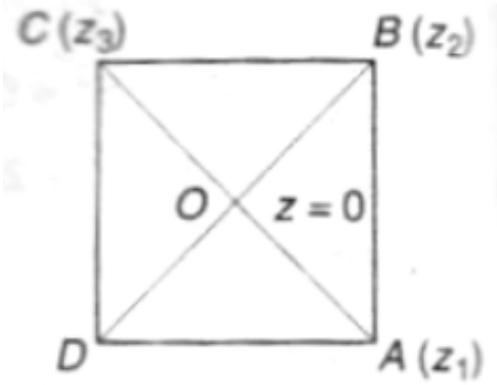
$(z_2 - 0) / (z_1 - 0) = i$

$z_2 = iz_1$

$\frac{z_3 - 0}{z_1 - 0} = \cos\pi + i\sin\pi \Rightarrow z_3 = -z_1$

$\text{centroid} = \frac{z_1 + z_2 + z_3}{3} = \frac{z_1(1 + i - 1)}{3} = i\frac{z_1}{3}$

@IITJEE_Advanced



Q.6 Answer:

is a hyperbola with its foci at $1-\mathrm{i}$ and $2+\mathrm{i}$ and its transverse axis $=3$

Solution:

Point (v) of passage: If major axis is to be considered as transversal axis with length $2\mathrm{a}<\left|\mathrm{z}_1-\mathrm{z}_2\right|$ then the equation of hyperbola is given by $\|z-z_1|-|z-z_2\|=2a$ where z_1 and z_2 are focii and transversal axis length as $2a$

Given equation: $|z-1+i|-|z-2-i|=3$

$\Rightarrow|z-(1-\mathrm{i})|-|z-((2+1)|=3$

This represents one half of a hyperbola with $(1-i)$ and $(2+i)$ as focii and transversal axis $=3$

Proof:

Let $z=x+iy$

$\Rightarrow|z-(1-i)|-|z-((2+i)|=3$

$\Rightarrow|x+iy-(1-i)|-|x+iy-((2+i)|=3$

$\Rightarrow\sqrt{(\mathrm{x}-1)^2+(\mathrm{y}+1)^2}-\sqrt{(\mathrm{x}-2)^2+(\mathrm{y}-1)^2}=3$

$\Rightarrow\sqrt{(\mathrm{x}-1)^2+(\mathrm{y}+1)^2}=3+\sqrt{(\mathrm{x}-2)^2+(\mathrm{y}-1)^2}$

$\Rightarrow(\mathrm{x}-1)^2+(\mathrm{y}+1)^2=9+(\mathrm{x}-2)^2+(\mathrm{y}-1)^2+18\sqrt{(\mathrm{x}-2)^2+(\mathrm{y}-1)^2}$

$\Rightarrow-6+\mathrm{x}+2\mathrm{y}=9\sqrt{(\mathrm{x}-2)^2+(\mathrm{y}-1)^2}$

Again take squares on both sides and after rearranging: Proceeding further we get equation of hyperbola

Q.7 Answer:

$\frac{3\sqrt{3}|z|^2}{2}$

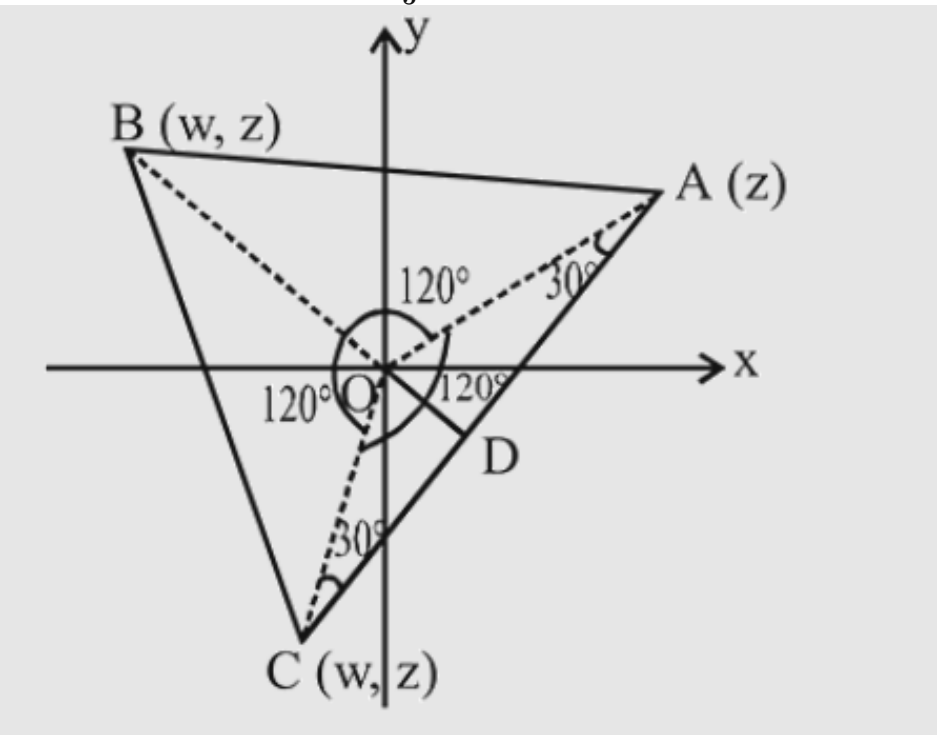
Solution:

Let the point A represents the complex number z, B represents $\omega \mathrm{z}$ and C represents $\bar{\omega} \mathrm{z}$

$\omega \& \bar{\omega}$ are complex cube roots of unity clearly

@IITJEE_Advanced

ωz means rotation of z by $\frac{2\pi}{3}$ and $\omega^2 z\ (=\bar{\omega} z)$ means rotation of ωz by $\frac{2\pi}{3}$.



$\therefore \angle AOB = \angle BOC = \angle COA = \frac{2\pi}{3}$ also
$OA = OB = OC = |z|$. That is the $\triangle ABC$ is equilateral.
Now $AC = 2AD = 2\,(OA \cos 30^\circ)$
$= 2|z|\frac{\sqrt{3}}{2} = \sqrt{3}|z|$
Area of $\Delta ABC = \frac{\sqrt{3}}{2}(\text{ side })^2 = \frac{3\sqrt{3}}{2}|z|^2$

Q.8 Answer:

$\frac{2}{3}$

Solution:

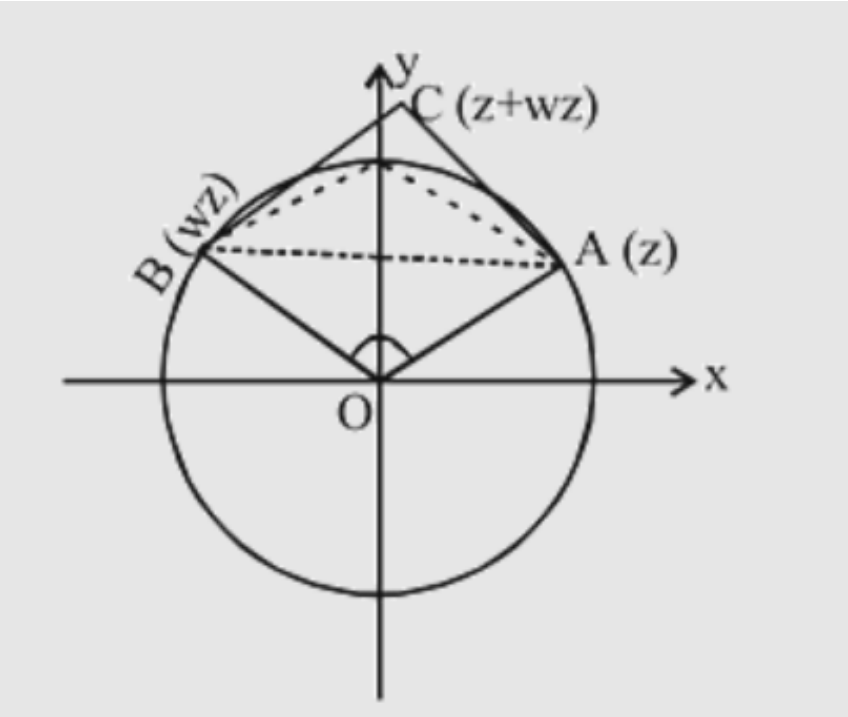
For the nonreal roots of the equation
$2z^2 + 2z + \lambda = 0 \quad \ldots$
discriminant < 0.
That is $4 - 8\lambda < 0 \Rightarrow \lambda > \frac{1}{2}$
Let the roots of (i) be $z_1 \& z_2$ Then $z_1 + z_2 = -\frac{2}{2} = -1, z_1 z_2 = \frac{\lambda}{2}$
z_1 and z_2 with origin form equilateral triangle if $z^2 + z_2^2 - z_1 z_2 = 0$
$\Rightarrow (z_1 + z_2)^2 = 3 z_1 z_2$
$\Rightarrow (-1)^2 = 3\frac{\lambda}{2} \Rightarrow \lambda = \frac{2}{3}$
$\lambda = \frac{2}{3}\left(> \frac{1}{2}\right)$ satisfies the condition (ii). Hence it is the required result.

Q.9 Answer:

4 units

Solution:

@IITJEE_Advanced



Let the point A represents z and B represents ωz, then $\angle \mathrm{AOB}=\frac{2\pi}{3}$. If C represents $\mathrm{z}+\omega \mathrm{z}$ then OACB is a
parallelogram. $\therefore$ Area of $\Delta \mathrm{ABC}=$ Area of $\Delta \mathrm{OAB}$
$=\frac{1}{2}|\mathrm{z}||\omega \mathrm{z}| \sin \frac{2\pi}{3}=\frac{1}{2}|\mathrm{z}||\mathrm{z}| \frac{\sqrt{3}}{2}[|\omega \mathrm{z}|=|\mathrm{z}|]=\frac{\sqrt{3}}{4}|\mathrm{z}|^2$
Now given that area of $\triangle \mathrm{ABC}=4\sqrt{3}$.
$\frac{\sqrt{3}}{4}|\mathrm{z}|^2=4\sqrt{3}$
$\Rightarrow |z|=4$. **That is z lies on a circle or radius 4**

Q.10 Answer:

6 and 0

Solution:

We have $|z+1|=|z+4-3|$ ...(i)
Now $|z+4-3| \leq |z+4|+|-3| \leq 3+3=6$
[Given $|z+4| \leq 3 \& |-3|=3]$ $\therefore |z+1| \leq 6$
Again $|z+1| \geq 0$ [modulus is always non-negative]
$\therefore$ Least value of $|\mathrm{z}+1|$ may be zero, which occurs when $\mathrm{z}=-1$, For $\mathrm{z}=-1$, $|\mathrm{z}+4|=|-1+4|=3$
Which satisfies the given condition that $|\mathrm{z}+4| \geq 3$ Hence, the least and the greatest values of $|z+1$ are 0 and 6

Q.11 Answer:

1

Solution:

@IITJEE_Advanced

$z_1 + z_2 = -1, z_1 z_2 = \frac{b}{3}$
$0^2 + z_1^2 + z_2^2 = 0 \times z_1 + 0 \times z_2 + z_1 z_2$
$\Rightarrow \quad (z_1 + z_2)^2 - 2z_1 z_2 = z_1 z_2 \Rightarrow 1 = 3z_1 z_2 = 3\frac{b}{3}$
$\Rightarrow b = 1$

Q.12 Answer:

3

Solution:

Let the vertices be $z_0, z_1, \ldots\ldots Z_5$ w.r.t centre O as origin $|z_0| = 1$,

$A_0\ A_1 = |z_1 - z_0| = \left|z_0 e^{i\theta} - z_0\right|$

$\therefore A_0\ A_1 = |z_0||\cos\theta + i\sin\theta - 1|$

$= 1 \cdot \sqrt{(\cos\theta - 1)^2 + \sin^2\theta} = \sqrt{2(1 - \cos\theta)}$

$\therefore A_0\ A_1 = \sqrt{2.2\sin^2\frac{\theta}{2}} = 2\sin\frac{\theta}{2}$

Where $\theta = \frac{2\pi}{6} = \frac{\pi}{3}$. Replacing θ by 2θ and 4θ we get, $A_0A_2 = 2\sin\frac{2\theta}{2} = 2\sin\theta$&

$A_0\ A_4 = 2\sin\frac{4\theta}{2} = 2\sin 2\theta$

$\therefore (A_0\ A_1)\,(A_0\ A_2)\,(A_0\ A_4)$

$= 8\sin\frac{\pi}{6}\sin\frac{\pi}{3}\sin\frac{2\pi}{3}$

$= 8\left(\frac{1}{2}\right)\left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{3}}{2}\right) = 3$

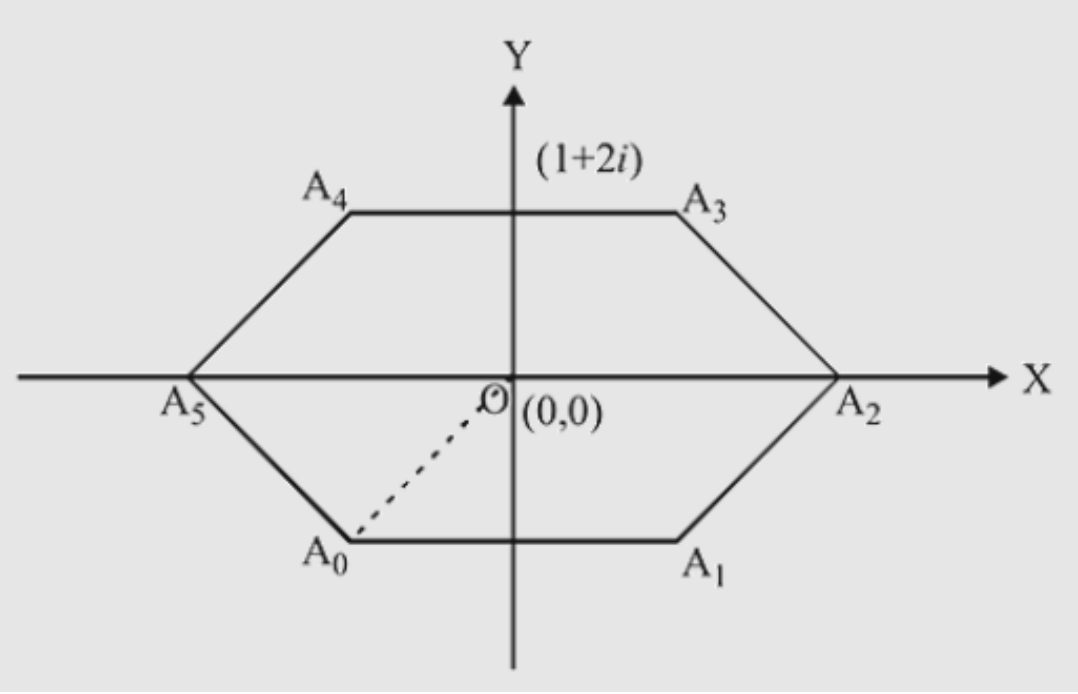
Q.13 Answer:

$6\sqrt{5}$

Solution:

Let the vertices be $z_0, z_1, \ldots, z_5$ w.r.t. centre O at origin and $|z_0| = \sqrt{5}$.

@IITJEE_Advanced



$\Rightarrow \quad \mathrm{A}_0\ \mathrm{A}_1 = |z_1 - z_0| = \left|z_0 e^{i\theta} - z_0\right|$
$= |z_0|\cos\theta + i\sin\theta - 1\ |$
$= \sqrt{5}\sqrt{(\cos\theta - 1)^2 + \sin^2\theta}$
$= \sqrt{5}\sqrt{2(1-\cos\theta)} = \sqrt{52}\sin(\theta/2)$
$\Rightarrow \mathrm{A}_0\ \mathrm{A}_1 = \sqrt{5}\cdot 2\sin\left(\frac{\pi}{6}\right) = \sqrt{5}$
$\left(\because \quad \theta = \frac{2\pi}{6} = \frac{\pi}{3}\right)$

Similarly, $\mathrm{A}_1\ \mathrm{A}_2 = \mathrm{A}_2\ \mathrm{A}_3 = \mathrm{A}_3\ \mathrm{A}_4 = \mathrm{A}_4\ \mathrm{A}_5 + \mathrm{A}_5\ \mathrm{A}_0 = 6\sqrt{5}.$

Hence the perimeter of, regular polygon is $= \mathrm{A}_0\ \mathrm{A}_1 + \mathrm{A}_1\ \mathrm{A}_2 + \mathrm{A}_2\ \mathrm{A}_3 + \mathrm{A}_3\ \mathrm{A}_4 + \mathrm{A}_4\ \mathrm{A}_5 + \mathrm{A}_5\ \mathrm{A}_0 = 6\sqrt{5}$

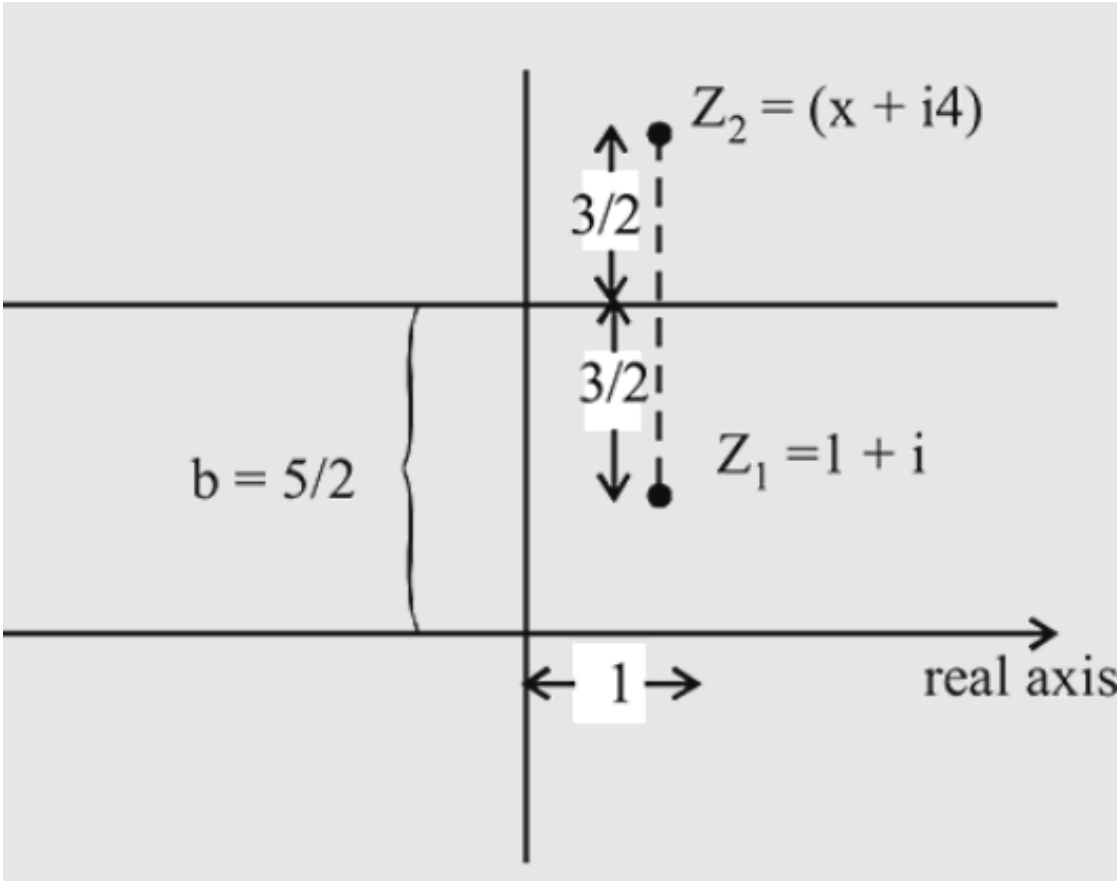
Q.14 Answer:

$1 + 4\mathrm{i}$

Solution:

@IITJEE_Advanced

Let $z = a + bi$



$\Rightarrow \bar{z} = a - bi$

$\therefore i\bar{z} - iz = i[(a - bi) - (a + bi)] = 5$

$\Rightarrow i[-2bi] = 5$

$\Rightarrow b = \frac{5}{2}$

So from figure it is clear that

$x = 1, y = \frac{5}{2} + \frac{3}{2} = 4$

$z_2 = 1 + 4i$

Q.15 Answer:

$\text{Re}(z) \in \{\frac{3}{2}, \frac{7}{2}\}$

Solution:

$|z - 2| = \min\{|z - 1|, |z - 5|\}$

i.e., $|z - 2| = |z - 1|$**, where** $|z - 1| < |z - 5|$

$\Rightarrow \quad \text{Re}(z) = \frac{3}{2}$ **which satisfy** $|z - 1| < |z - 5|$

Also, $|z - 2| = |z - 5|$**, where** $|z - 5| < |z - 1|$

$\Rightarrow \quad \text{Re}(z) = \frac{7}{2}$ **which satisfy** $|z - 5| < |z - 1|$

@IITJEE_Advanced